

2. LOW-LEVEL DESIGN (LLD)

1. Module-Level Design

- Document Processing Module
 - Chunking Module
 - Embedding Module
 - Vector Storage Module
 - Retrieval Module
 - Query Processing Module
 - Graph Execution Module
 - HITL Module
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2. Data Structures

- **Document:** Raw text from PDF
- **Chunk:**

```
{  
  text: str,  
  metadata: {page_number}  
}
```

- **Embedding:** Vector representation
- **Query-Response Schema:**

```
{  
  query: str,  
  context: list,  
  response: str  
}
```

- **Graph State:**

```
{  
  query,  
  retrieved_docs,
```

```
    response,  
    route  
}
```

3. Workflow Design (LangGraph)

- **Nodes:**
 - Processing Node
 - Output Node
 - **Edges:**
 - Input → Processing
 - Processing → Output / HITL
 - **State:**
 - Carries query, docs, response
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4. Conditional Routing Logic

Current Implementation

- No documents → HITL
- Few documents → HITL

Improved Logic (Conceptual Upgrade)

- Low retrieval confidence → HITL
 - Ambiguous query → HITL
 - Complex query → HITL
 - Otherwise → Answer
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5. HITL Design

- Triggered when:
 - No relevant context
 - Low confidence
 - System response:
 - Escalates query
 - Human provides answer
 - Integration:
 - Human response replaces LLM output
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6. API / Interface Design

Input

```
{  
  "query": "string"  
}
```

Output

```
{  
  "response": "string",  
  "source": "LLM | HITL"  
}
```

7. Error Handling

- Missing data → fallback response
- No chunks → escalate
- LLM failure → retry / fallback
- Invalid query → validation